

Adapting Pretrained Language Models for Citation Classification via Self-Supervised Contrastive Learning

Code: github.com/LITONG99/Citss
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Background & Challenges

Citation Classification: *why authors cite other papers?*

The textual context surrounding the citation anchor is essential to reveal the intention. Pretrained language models (PLMs) can be fine-tuned to discern citation intentions. However, the fine-tuning confronts unique challenges.

Challenge 1. Scarcity of labeled citation data.

It requires domain-specific knowledge and expertise of annotators to interpret the scientific texts and assign labels. Existing works call on author self-annotation or employ annotators with academic backgrounds. Collected datasets remain limited to a few thousand samples, which fall short in manifesting the task-specific patterns and restrain the performance of PLMs.

Challenge 2. Defocusing on the target citation.

Most methods choose to input citance, the citing sentence, which is (1) highly related, (2) easy for feature design, (3) but not including long-range dependencies. PLMs can use a long context as input, which (1) includes more information, but also (2) introduce excessive irrelevant descriptions, such as mentions of other citations and general discourses.

Challenge 3. Spurious correlation based on keyphrases.

The name of research subjects, tasks, techniques, materials, etc. are (1) repeatedly mentioned and (2) semantically significant during pretraining, (3) usually not indicative for any intention labels. PLMs can easily establish toxic correlations from STKs in the text to the label during fine-tuning.

target citation citance

.....As computer vision techniques have improved over the past decade, other research has begun directly using visual information in place of feature norms. The first work to do this with topic models is [1]. They use a Bag of Visual Words model to create a bimodal vocabulary describing documents. The topic model using the bimodal vocabulary outperforms a purely textual based model in word association and word similarity prediction. Bruni et al. [2] show how a BoVW model may be easily combined with a distributional vector space model of language....

long context other citations

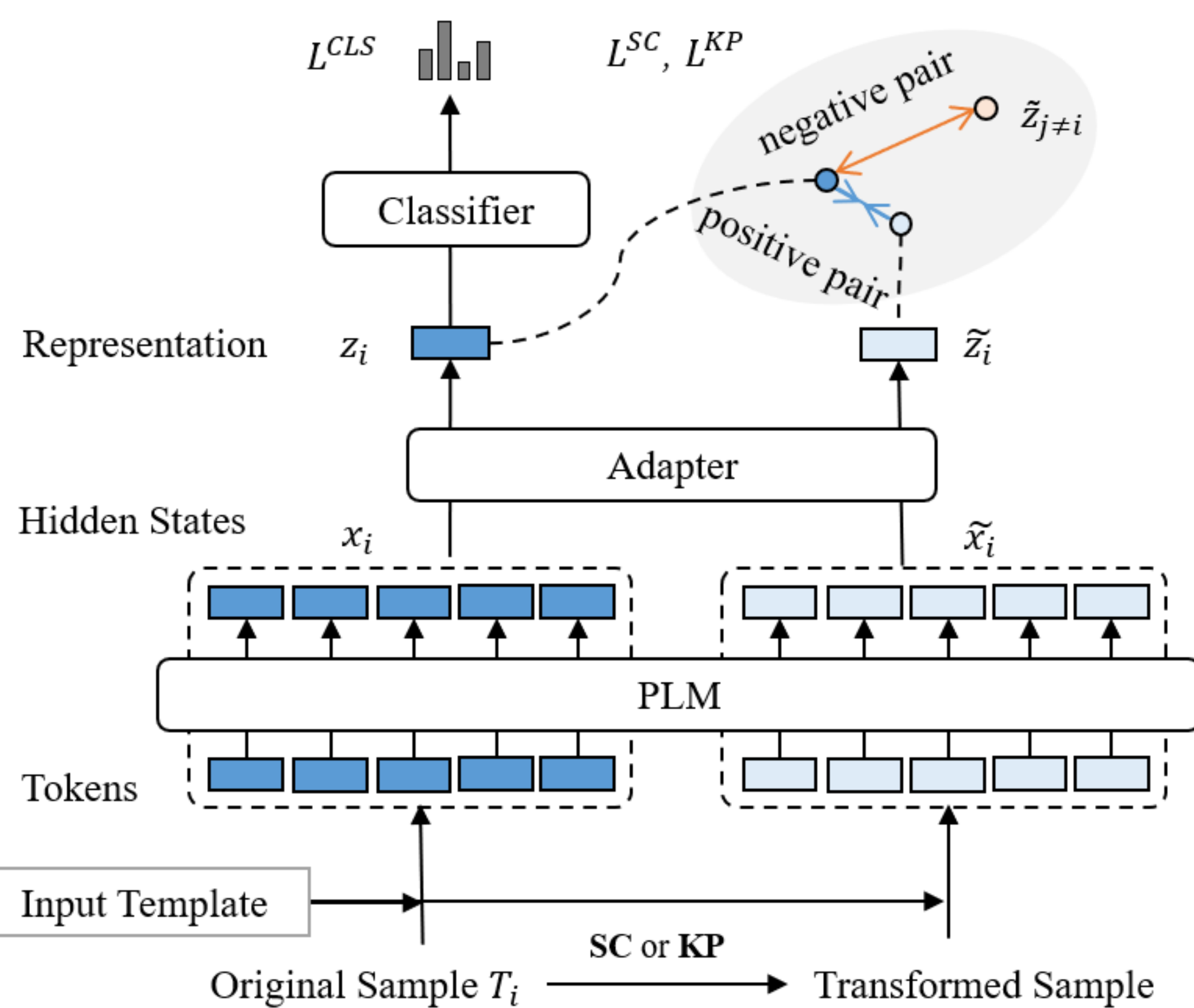
Generate contrastive pairs in a self-supervised manner and derive contrastive loss to provide extra supervision signals.

Sentence-level cropping strategy randomly crops a subsequence with the same citance but different context ranges, to enhance on focus on target citation.

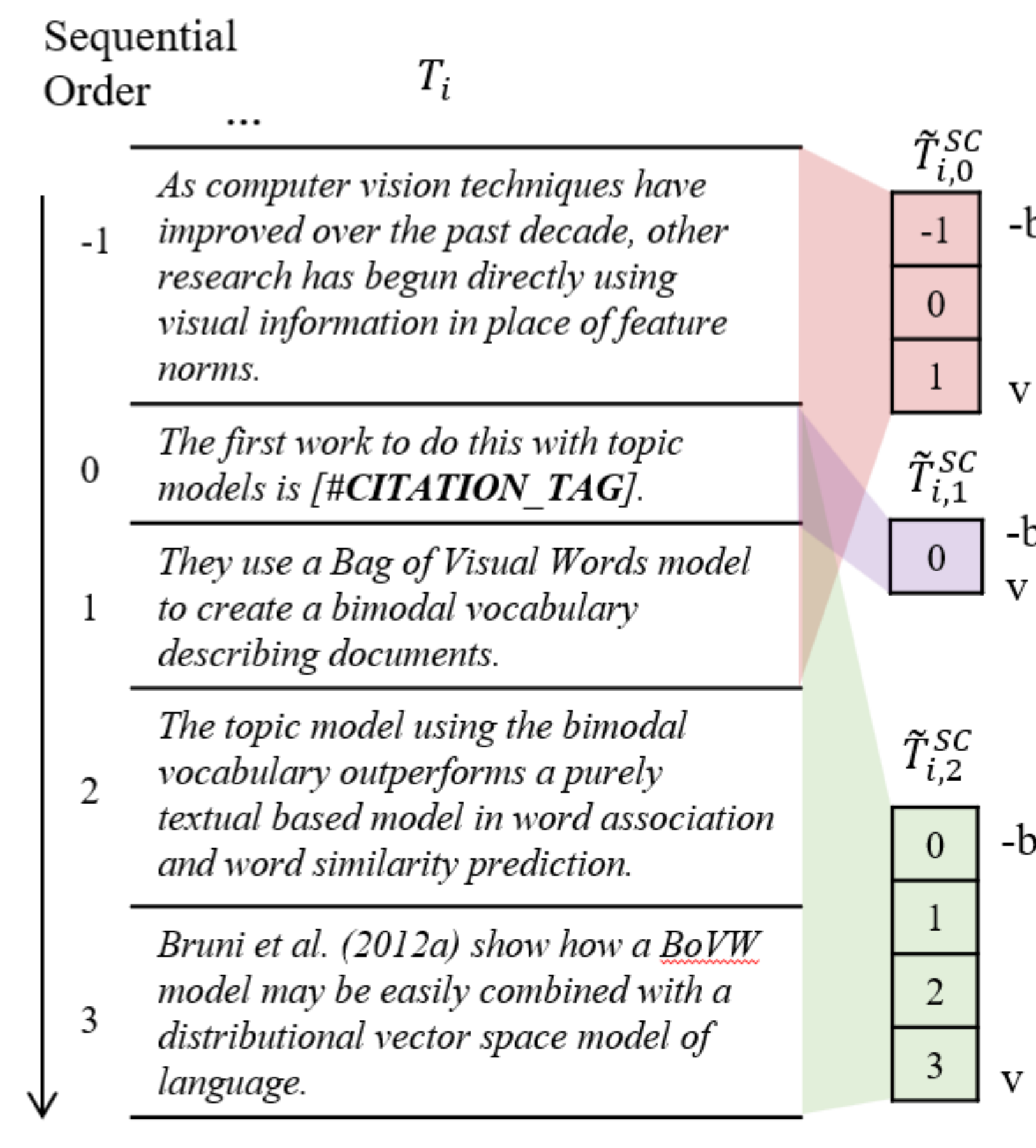
Modifying the STKs usually does not affect the citation intention. We design the keyphrase perturbation strategy to reduce the reliance on STKs.

Method

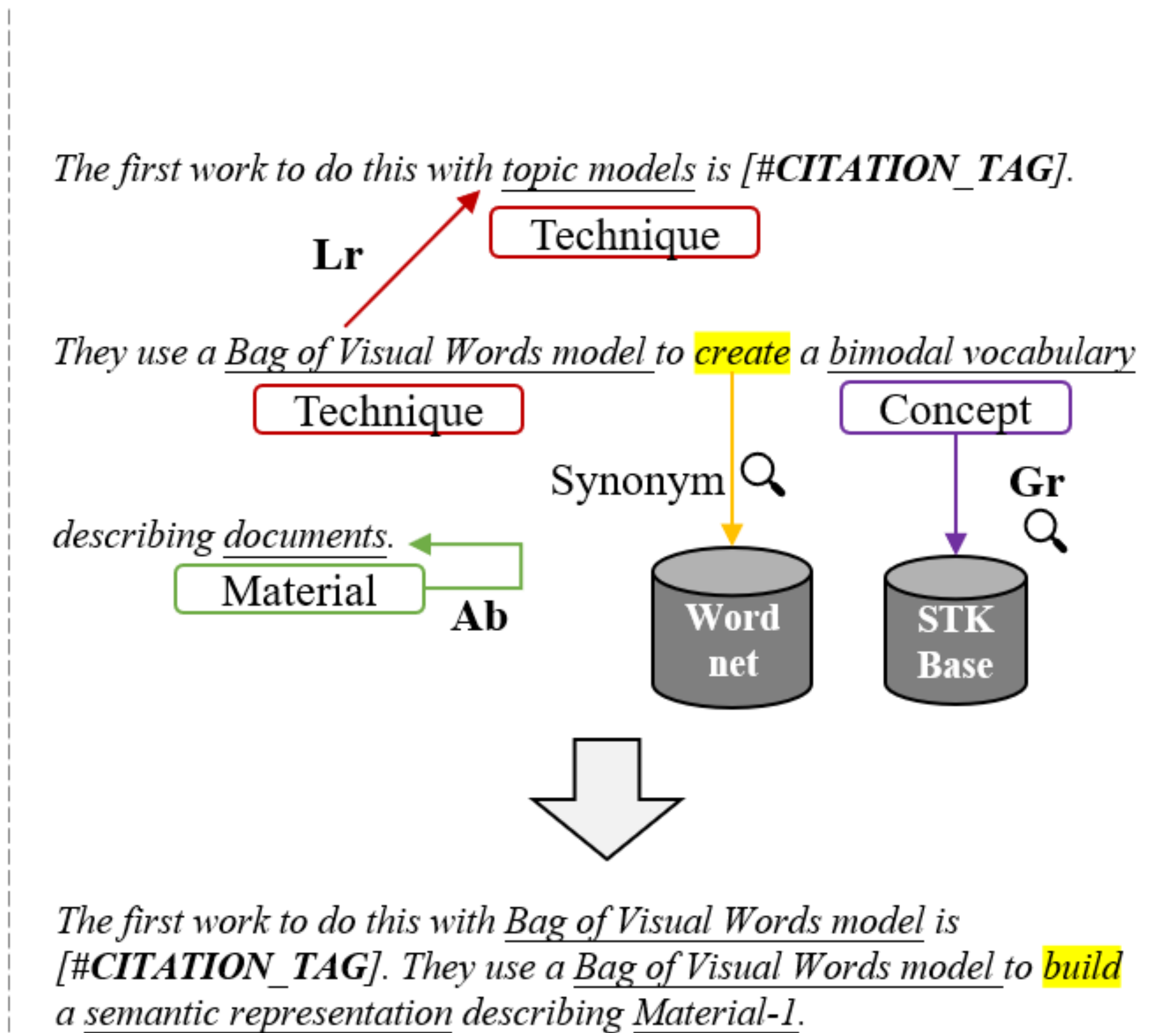
(a) Framework Architecture



(b) Sentence-Level Cropping (SC)



(c) Keyphrase Perturbation (KP)



Experiments

Overall Performance

We compare with our method **Citss** with fine-tuning and in-context learning baselines.

Citss can effectively fine-tune LLMs with extremely limited citation data.

Ablation Study

Method	Backbone	ACL-ARC		FOCAL		ACT2	
		Macro-f1	Accuracy	Macro-f1	Accuracy	Macro-f1	Accuracy
Scaffold	NA	0.496 ± 0.021	0.649 ± 0.012	0.145 ± 0.041	0.447 ± 0.133	0.146 ± 0.006	0.363 ± 0.017
IREL	SciBERT	0.614 ± 0.037	0.721 ± 0.025	0.580 ± 0.086	0.742 ± 0.007	0.262 ± 0.012	0.468 ± 0.028
TRL	SciBERT	0.476 ± 0.024	0.610 ± 0.007	0.604 ± 0.009	<u>0.756 ± 0.009</u>	0.118 ± 0.001	<u>0.544 ± 0.000</u>
PET	SciBERT	<u>0.616 ± 0.022</u>	<u>0.723 ± 0.019</u>	<u>0.641 ± 0.044</u>	0.750 ± 0.008	<u>0.258 ± 0.018</u>	0.537 ± 0.020
Citss	SciBERT	0.665* ± 0.018	0.743 ± 0.006	0.679 ± 0.023	0.777* ± 0.005	0.254 ± 0.012	0.563* ± 0.009
IFP	Llama3-8B	0.422	0.575	0.243	0.398	0.213	0.446
LoRA	Llama3-8B-bfloat16	<u>0.670 ± 0.050</u>	0.745 ± 0.028	0.670 ± 0.035	0.757 ± 0.001	0.242 ± 0.034	<u>0.529 ± 0.014</u>
Citss + LoRA	Llama3-8B-bfloat16	0.744 ± 0.010	0.819* ± 0.007	0.682 ± 0.024	0.768* ± 0.001	0.266 ± 0.006	0.549 ± 0.017
IFP	Llama3-70B	0.569	0.701	0.430	0.623	0.242	0.545

SciBERT	ACL-ARC			FOCAL			ACT2		
	Macro-f1	Accuracy	Imp.(%)	Macro-f1	Accuracy	Imp.(%)	Macro-f1	Accuracy	Imp.(%)
$\lambda_1, \lambda_2 = 0$	0.616 ± 0.012	0.714 ± 0.022	-	0.641 ± 0.029	0.742 ± 0.004	-	0.262 ± 0.016	0.539 ± 0.021	-
$\lambda_2 = 0$	0.660 ± 0.025	0.745 ± 0.032	5.7	0.666 ± 0.029	0.762 ± 0.006	3.3	0.246 ± 0.036	0.558 ± 0.008	0.3
$\lambda_1 = 0$	0.646 ± 0.032	0.730 ± 0.017	3.5	0.654 ± 0.004	0.764 ± 0.009	2.5	0.260 ± 0.003	0.549 ± 0.017	0.9
Llama3-8B	ACL-ARC			FOCAL			ACT2		
	Macro-f1	Accuracy	Imp.(%)	Macro-f1	Accuracy	Imp.(%)	Macro-f1	Accuracy	Imp.(%)
$\lambda_1, \lambda_2 = 0$	0.729 ± 0.007	0.799 ± 0.003	-	0.673 ± 0.008	0.762 ± 0.003	-	0.227 ± 0.006	0.544 ± 0.017	-
$\lambda_2 = 0$	0.736 ± 0.021	0.805 ± 0.013	0.9	0.681 ± 0.009	0.769 ± 0.004	1.0	0.248 ± 0.016	0.542 ± 0.016	2.5
$\lambda_1 = 0$	0.739 ± 0.008	0.804 ± 0.011	1.0	0.675 ± 0.005	0.766 ± 0.004	0.4	0.236 ± 0.009	0.556 ± 0.019	2.8

Our Paper

